

CONSTRUCTION NOTES FOR 10 PURPLE MARTIN NEST BOXES (TUBES)

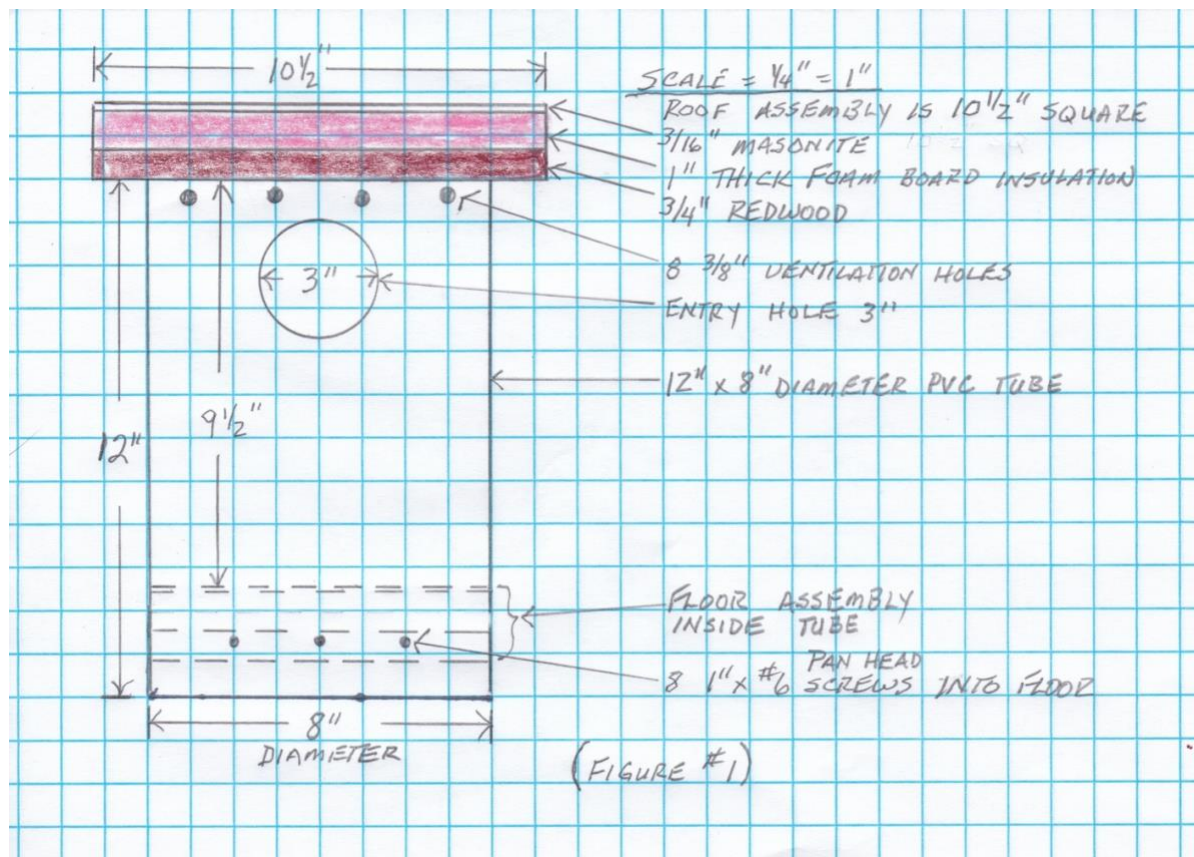


Figure #1

Description: This design uses a 12" long 8" diameter white PVC pipe lined with a 9 ½" long 6" diameter cement tube for the inside of the nest. Foam between the outer PVC pipe and inner cement tube insulates the box. The roof and floor use redwood fencing, 1" foam insulation, and 3/16" white masonite sandwiched together. The roof is 10 ½" square and the floor is a circle cut to fit inside the 8" diameter PVC tube. The nest box is designed to be mounted on top of a 1" galvanized pipe cut to desired length using a galvanized metal flange on the floor of the box. The nest boxes should last 15 to 25 years due to the plastic PVC outer shell and white Masonite roof.

Dimensions: 13 ¾" tall by 8" wide at the tube. The roof assembly is 10 ½" square.

Cost breakdown: less than \$50/each provided 10 nests are built at the same time to make best use of the material.

Insulation: The nest is insulated on all sides by 1" foam and/or spray foam insulation with a rating of R5. The entire outside of the nest is white to reduce temperature fluctuation.

Materials list:

1 ea	10' by 8" diameter white PVC pipe cut in to 10 ea. 12" lengths	\$55
2 ea	4' by 7" diameter Quik-krete cement tubes	\$24
6 ea	¾" by 5 ½" by 6' redwood fencing	\$36
3 ea	2' by 4' by 3/16" white Masonite markerboard	\$27
5 ea	2' by 2' pink foamular 1" think insulation	\$50
1 ea	Tube of liquid nails	\$ 5
20 ea	1 ½" by 1 ½" corner brackets	\$20
2 ea	Cans of spray insulation	\$15
80 ea	1" by #6 pan head screws	\$12
40 ea	1 ¾" by #6 pan head screws	\$10
10 ea	1" galvanized floor flange for mounting	\$150
1 ea	package of zip ties	\$ 5
Total		\$409

These costs do not include tax or shipping of the 10' PVC pipe. All other material may be purchased at any home store. See pictures of materials at the end of the design.

CONSTRUCTION:

Roof and Floor Board Preparation: The redwood fencing must be joined together to create boards wide enough for the roofs and floors. Cut the redwood fencing into 20 boards 8 ½" long and 20 boards 10 ¾" long. Rip both sides of all 40 boards to 5 ¼" width to provide a straight and square edge surface to install biscuits. Glue and clamp the 40 boards with wood biscuits to create 10 floor boards 8 ½"x10 ½" and 10 roof boards of 10 ¾" x 10 ½".

Roof Assembly: Cut the roof boards to a final size of 10 ½" square. Also cut 10 ½"squares of the foamular insulation and Masonite. Sandwich the three layers together using liquid nails with the white side of the Masonite on top, the foam in the middle, and the redwood board on the bottom. Secure together with a 1 ¾" screw in each corner. See Figure #1 and Figure #2.

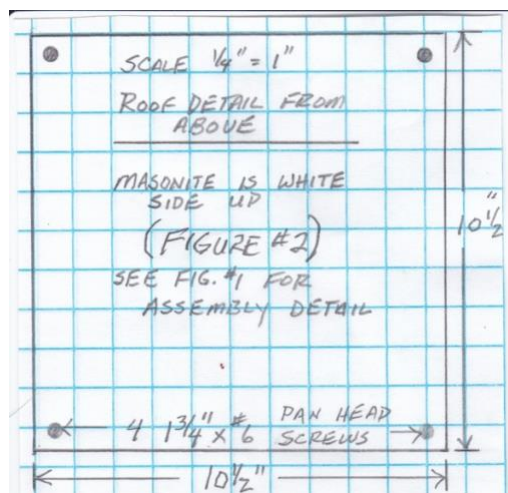


Figure #2

Floor Assembly: With a pencil, use a PVC tube to trace the inside diameter onto the floor boards, foamular and Masonite. Cut the round floorboards and Masonite with a jigsaw. Cut the foamular with a utility knife. Sandwich the three boards together with liquid nails. Drill eight $\frac{3}{8}$ " drainage holes $1\frac{1}{2}$ " in from outer edge as depicted in diagram. Center and mount the 1" galvanized floor flange on the bottom into the redwood. See Figure #3.

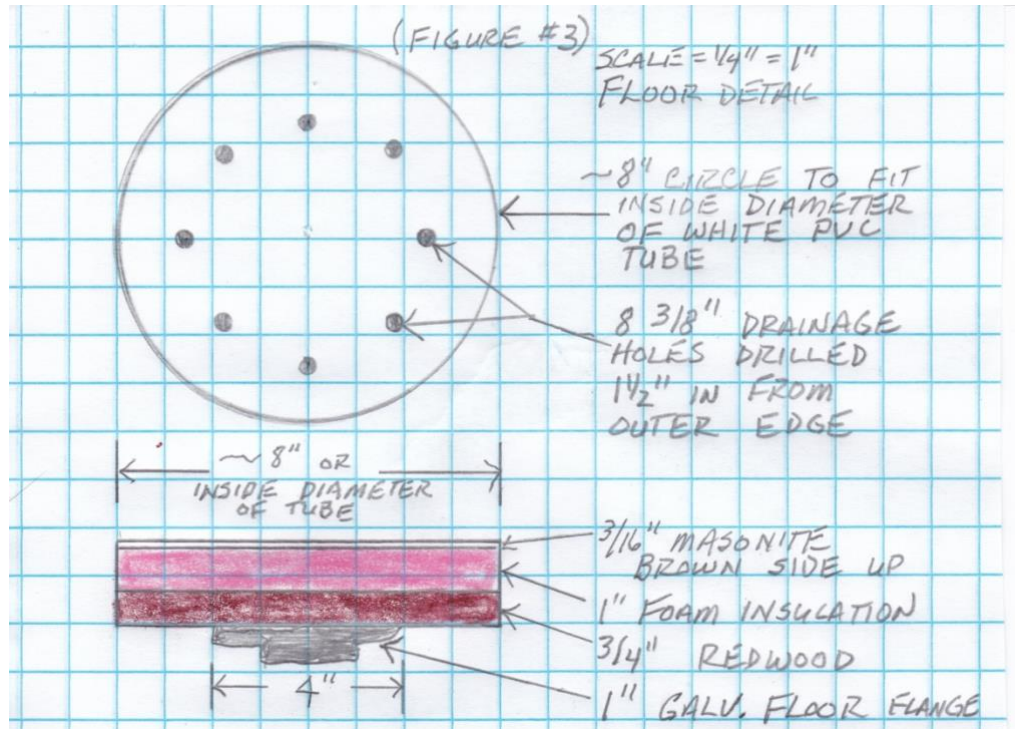


Figure #3

Mount floor assembly inside PVC tube using 8 1" by #6 pan head screws. Locate the center of where the redwood portion of floor assembly is and predrill 8 holes before securing the floor inside the PVC with 1" screws. See Figure #1 for detail.

Nest Box Liner Assembly: Take the 7" diameter by 9 1/2" long cement tube and cut a 3 1/2" strip wide strip from the length of the tube. Locate area where entry hole will be cut and etch the inside of cement tube below entry hole for fledglings to crawl up to hole. (Note: without actually constructing this, I am not sure what tool(s) to use for this part of construction.) Using zip ties on both ends of the cement tube, cinch the tube down to form a 6" diameter tube. See design below in Figure #4.

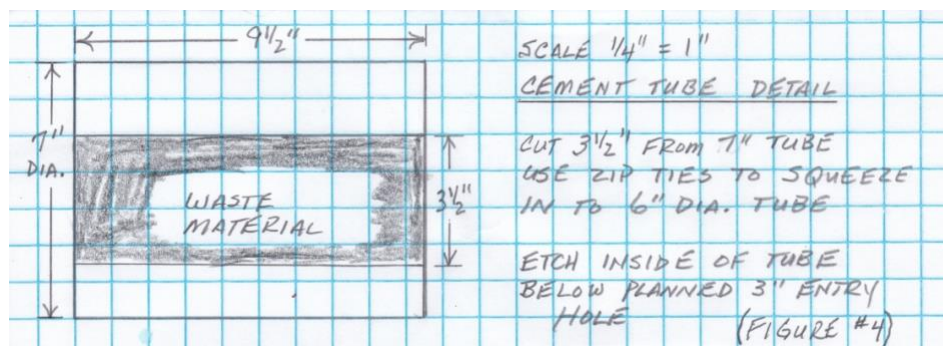


Figure #4

Center the 6" cement tube on top of floor assembly inside the 8" PVC tube. While holding securely in place to the floor, spray a small amount of spray foam into bottom to secure cement tube in place. Allow foam spray to set before proceeding. Cut leftover pieces of pink foam insulation into 1" square chunks and use these along with the spray foam to fill the 1" gap between PVC and cement tubes. Allow to cure. Cut any foam that may have risen above the top of the tubes to provide a flat surface for mounting the roof. See tube detail below.

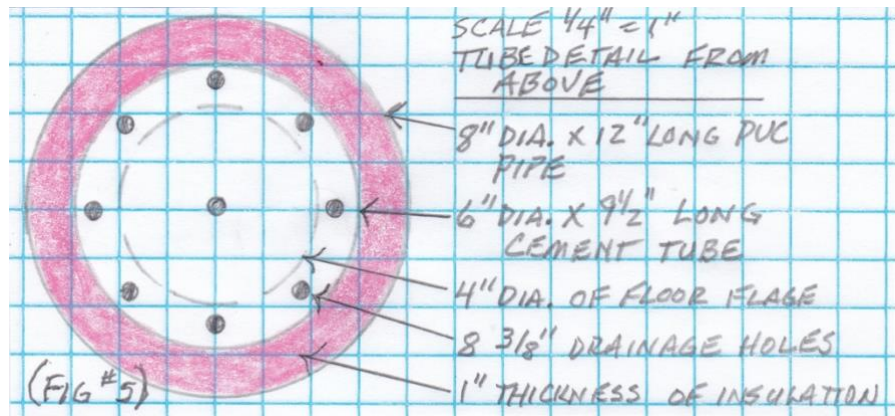


Figure #5

Entry Hole Construction: Drill a $\frac{1}{4}$ " pilot hole $2\frac{1}{2}$ " down from the top and over the area previously etched for the entry hole. Use a 3" hole saw, drill through the PVC and into the insulation as far as possible. Remove the saw and clean out the PVC and any insulation. Still using the pilot hole, drill the remaining 3" hole through the cement tube. This puts the entry hole 1" down from the roof.

Drill eight $\frac{3}{8}$ " holes $\frac{1}{2}$ " down from top around the circumference to provide ventilation.

Roof Mounting: Mount roof to the top of the PVC tube by predrilling the PVC and using $1\frac{1}{2}$ " by $1\frac{1}{2}$ " corner brackets screwed into the redwood roof assembly. This would be the way to remove the top for cleaning of the nest, if needed.

Following are pictures of the some of the needed material.

Redwood fence material $\frac{3}{4}$ " by 5 $\frac{1}{2}$ " by 6'



Masonite markerboard $\frac{3}{16}$ " by 2' by 4'



Cement Tubes



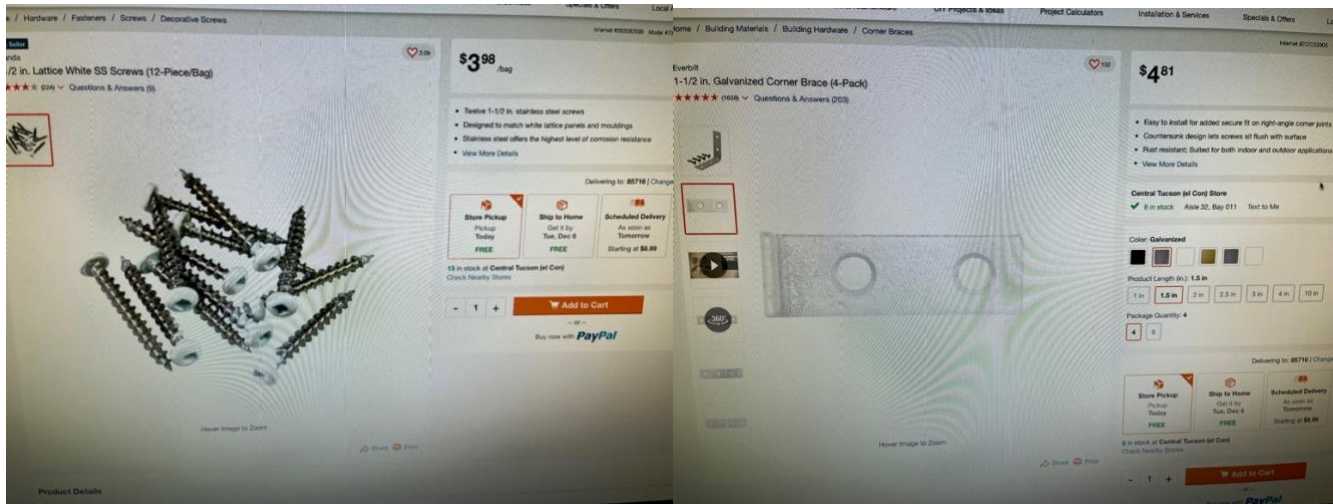
Foamular 1" Foam Insulation



Liquid Nails

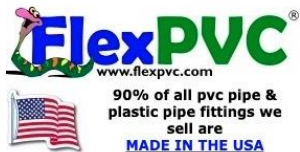


Spray Foam Insulation



Pan Head Screws

Corner Brackets



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CL 100 aka SDR41 8" UltraThin Wall PVC Pipe, SEE DETAILS



8" SDR41 aka, Class 100 8" PVC Pipe ID=8.205" Min. Wall thickness: 0.210". Fits standard Sch 40/80 fittings, but ultra thin wall allows maximum flow with very small weight. Max pressure 100PSI. May not be available all the time. Stock fluctuates widely. Note: Length must be between 6" and 72". If you specify anything longer it could result in additional shipping and handling fees being added to your order. For pieces less than 6" or longer than 72" please email us for a quote)

Price: \$53.35

1" Threaded Galvanized Floor Flange

